Kinetic Theory (Supplementary Problems)

1. Ordinary nitrogen gas consists of molecules of N_2. Find the mass of one such molecule. The molecular mass is 28 kg/kmol.

2. Helium gas consists of separate He atoms rather than molecules. How many helium atoms, He, are there in 2.0 g of helium? M = 4.0 kg/kmol for He.

3. A droplet of mercury has a radius of 0.50 mm. How many mercury atoms are in the droplet? For Hg, M = 202 kg/kmol and $\rho = 13\ 600\ kg/m^3$.

4. How many molecules are there in 70 mL of benzene? For benzene, $\rho = 0.88\ g/cm^3$ and M = 78 kg/kmol.

5. Find the rms speed of a nitrogen molecule (M = 28 kg/kmol) in air at 0 °C.

6. A gas molecule at the surface of the Earth happens to have the rms speed for that gas at exactly 0 °C. If it were to go straight up without colliding with other molecules, how high would it rise?

7. Air at room temperature has a density of about 1.29 kg/m3. Assuming it to be entirely one gas, find n_{rms} for its molecules.

8. Find the translational kinetic energy of one gram mole of any ideal gas at 0 °C.

9. There is about one hydrogen atom per cm^3 in outer space, where the temperature (in the shade) is about 3.5 K. Find the rms speed of these atoms and the pressure they exert.

10. Find the following ratios for hydrogen (M = 2.0 kg/kmol) and nitrogen (M = 28 kg/kmol) gases at the same temperature: (a) $(KE)_H/(KE)_N$ and (b) $(\text{rms speed})_H/(\text{rms speed})_N$.

11. Find the mass of a neon atom. The atomic mass of neon is 20.2 kg/kmol. Ans. 3.36 × 10—26 kg

12. A typical polymer molecule in polyethylene might have a molecular mass of 15×10^3. (a) What is the mass in kilograms of such a molecule? (b) How many such molecules would make up 2 g of polymer? Ans. (a) 2.5×10^{23} kg; (b) 8×10^{19}

13. A certain strain of tobacco mosaic virus has $M = 4.0 \times 10^7$ kg/kmol. How many molecules of the virus are present in 1.0 mL of a solution that contains 0.10 mg of virus per mL? Ans. 1.5×10

14. An electronic vacuum tube was sealed off during manufacture at a pressure of 1.2 10—7 mmHg at 27 °C. Its volume is 100 cm3. (a) What is the pressure in the tube (in Pa)? (b) How many gas molecules remain in the tube? Ans. (a) 1.6×10^5 Pa; (b) 3.8×10^{11}

15. The pressure of helium gas in a tube is 0.200 mmHg. If the temperature of the gas is 20 °C, what is the density of the gas? (Use MHe = 4.0 kg/kmol.) Ans. 4.4×10^5 kg/m^3

16. At what temperature will the molecules of an ideal gas have twice the rms speed they have at 20 °C? Ans. 1170 K ≈ 900 °C

17. An object must have a speed of at least 11.2 km/s to escape from the Earth‘s gravitational field. At what temperature will nrms for H_2 molecules equal the escape speed? Repeat for N_2 molecules. (MH2 = 2.0 kg/ kmol and MN2 = 28 kg/kmol.) Ans. 1.0 × 10 K; 1.4 × 10 K

18. In a certain region of outer space there are an average of only five molecules per cm3. The temperature there is about 3 K. What is the average pressure of this very dilute gas? Ans. 2×10^{16} Pa

19. A cube of aluminum has a volume of 1.0 cm3 and a mass of 2.7 g. (a) How many aluminum atoms are there in the cube? (b) How large a volume is associated with each atom? (c) If each atom were a cube, what would be its edge length? M = 108 kg/kmol for aluminum. Ans. (a) 1.5 × 1022; (b) 6.6 × 10—29 m3; (c) 4.0×10^{10} m

20. The rms speed of nitrogen molecules in the air at S.T.P. is about 490 m/s. Find their mean free path and the average time between collisions. The radius of a nitrogen molecule can be taken to be $2.0 \times 10^{-}10$ m. Ans. 5.2×10^{8} m, 1.1×10^{10} s

21. What is the mean free path of a gas molecule (radius 2.5 × 10—10 m) in an ideal gas at 500 °C when the pressure is 7.0×10^{6} mmHg? Ans. 10 m

22. a A sample of an ideal monatomic gas in a closed container at 18 °C has a total internal energy of 380 J. Determine how many particles (atoms) it contains.
b If the container has a volume of 435 cm3, calculate the pressure of the gas.

23. a Calculate the average random translational kinetic energy of oxygen (O_2) molecules at 300 K.
b Oxygen molecules each have a mass of 5.31×10^{-26} kg. Determine their average speed at 300 K.
c Explain why an average carbon dioxide (CO_2) molecule travels slower than oxygen molecules (O_2) at the same temperature.

24. a Determine how much energy is needed to raise the temperature of two moles of a monatomic gas from 20 °C to 50 °C.
b Why will more energy be needed to raise two moles of a *molecular* gas through the same temperature rise?

25. At what temperature will 1.0×10^{23} atoms of an ideal gas have a total internal energy of 1000 J?

26. Some nitrogen gas was cooled from a temperature of 300 K to 100 K.

a Estimate by what percentage its pressure was reduced.

b State two assumptions you had to make in order to answer part **a**.

c Explain why the actual final pressure was less than that predicted by theory.

27. What is the pressure of an ideal gas which has a temperature of 47 °C and contains 4.2×10^{25} particles in a volume of 0.037 m^3?

28. a Explain why the internal energy of an ideal gas can be determined from: 1.5 × pressure × volume.
b Discuss why temperature does not appear in this equation.
c What volume (cm3) of an ideal gas has an internal energy of 10 J and exerts a pressure of twice normal atmospheric pressure?

29. Here are five numbers: 5, 11, 32, 67, and 89.

(a) What is the average value n_{avg} of these numbers?

(b) What is the rms value n_{rms} of these numbers?

30. Nine particles have speeds of 5.00, 8.00, 12.0, 12.0, 12.0, 14.0, 14.0, 17.0, and 20.0 m/s.

a). Find the particles' average speed.

b). What is the rms speed of the particles?

31. Eight molecules have speeds of 3.00 km/s, 4.00 km/s, 5.80 km/s, 2.50 km/s, 3.60 km/s, 1.90 km/s, 3.80 km/s, and 6.60 km/s. Find (a) the average speed of the molecules and (b) the rms speed of the molecules.

32. Fifteen identical particles have various speeds: one has a speed of 2.00 m/s, two have speeds of 3.00 m/s, three have speeds of 5.00 m/s, four have speeds of 7.00 m/s, three have speeds of 9.00 m/s, and two have speeds of 12.0 m/s. Find (a) the average speed, (b) the rms speed, and (c) the most probable speed of these particles. **Ans: a).** $\bar{v}$ **= 6.80 m/s, b).** v_{rms} **= 7.41m/s**

33. A cylinder contains a mixture of helium and argon gas in equilibrium at 150°C. (a) What is the average kinetic energy for each type of gas molecule? (b) What is the rms speed of each type of molecule?
Ans: a). $K_{av} = 8.76 \times 10^{-21}$ J b). v = 514 m/s

34. (a) How many atoms of helium gas fill a spherical balloon of diameter 30.0 cm at 20.0°C and 1.00 atm? (b) What is the average kinetic energy of the helium atoms? (c) What is the rms speed of the helium atoms?
Ans: a). $N = 3.54 \times 1023$ **helium atoms, b).** $K_{av} = 6.07 \times 10^{-21}$ **J, c).** $v_{rms} = 1.35$ **km/s**

35. Eleven molecules have speeds 15, 16, 17, . . ., 25 m/s. Calculate (a) $\bar{v}$ and (b) v_{rms}.

36. The molecules in a six-particle gas have velocities

$\vec{v}_1 = \left(20\hat{i} - 30\,j\right) m/s$ $\quad$ $\vec{v}_4 = 30\hat{i}\, m/s$

$\vec{v}_2 = \left(40\hat{i} + 70\,j\right) m/s$ $\quad$ $\vec{v}_5 = \left(40\hat{i} - 40\,j\right) m/s$

$\vec{v}_3 = \left(-80\hat{i} + 20\,j\right) m/s$ $\quad$ $\vec{v}_6 = \left(-50\hat{i} - 20\,j\right) m/s$

Calculate (a) $\vec{\bar{v}}$, (b) $\bar{v}$ and (b) v_{rms}.

37. (a) What is the average translational kinetic energy of a molecule of oxygen (O_2) at a temperature of 27°C, assuming that oxygen can be treated as an ideal gas? (b) What is the total translational kinetic energy of the molecules in 1 mole of oxygen at this temperature? (c) Compare the root-mean-square speeds of oxygen (O_2) and nitrogen (N_2) molecules at this temperature, assuming they can also be treated as ideal gases. The mass of an oxygen molecule is $m_{O_2} = 5.31\times10^{-26}$ kg, and the mass of a nitrogen molecule is $m_{N_2} = 4.65\times10^{-26}$ kg.

38. Five ideal-gas molecules chosen at random are found to have speeds of 500, 600, 700, 800, and respectively. Find the rms speed for this collection. Is it the same as the *average* speed of these molecules? 900 m/s,

39. The speed of group of 10 molecules are 2,3,4,5,6,7,8,9,10,11 km/s. (a) Find the average speed of group. (b) Calculate the root mean square speed of the group.

40. The speeds of 22 particles are as follows (N_i represents the number of particles that have speed v_i):

N_i	2	4	6	8	2
v_i (cm/s)	1.0	2.0	3.0	4.0	5.0

What are (a) v_{avg}, (b) v_{rms},?

41. Ten particles are moving with the following speeds: four at 200 m/s, two at 500 m/s, and four at 600 m/s. Calculate their (a) average and (b) rms speeds. (c) Is $v_{rms} > v_{avg}$?

42. The average value of the squared speed $\overline{v^2}$ does not equal the square of the average speed $\left(\bar{v}\right)^2$. To verify this fact, consider three particles with the following speeds: v_1 = 3.0 m/s, v_2 = 7.0 m/s, and v_3 = 9.0 m/s. Calculate: a. $\overline{v^2} = \frac{1}{3}\left(v_1^2 + v_2^2 + v_3^3\right)$ and b. $\left(\bar{v}\right)^2 = \left[\frac{1}{3}\left(v_1 + v_2 + v_3\right)\right]^2$